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- ① rigid body kinematics
- ② rigid body kinetics : 힘이 가해졌을 때 어떻게 운동할 것인가?
- ③ control : 특정 자세로 어떻게 돌아갈 것인가?

* Kinematics

- 1. point mass
- 2. rigid body $\rightarrow$ cosine matrix, rotation matrix, $\oplus$ classic 한 yaw, pitch, roll rotation vector. quaternions?
no singularity

* Particle Kinematics

$$r = x\hat{e}_1 + y\hat{e}_2 + z\hat{e}_3$$

$$= r\hat{e}_r$$

$$= \begin{pmatrix} x \\ y \\ z \end{pmatrix}$$

$\rightarrow$ vector를 matrix 형태로 표현한 것

$\rightarrow$ 모든 vector는 matrix로 나타낼 수 있지만 모든 matrix는 vector로 나타낼 수는 없다.

$\rightarrow$ 매 자체는 좌표계가 정해지지 않은 vector로 나타낼 수 있음

여기서 vector의 의미에 대해 생각해야 한다. $\vec{q} = \vec{r} + \vec{p}$ 라는 식에서 r, p, q는 같은 frame 안에 있지 않은 수도 있다.

$${}^E \begin{pmatrix} q_1 \\ q_2 \\ q_3 \end{pmatrix} = {}^E \begin{pmatrix} r_1 \\ r_2 \\ r_3 \end{pmatrix} + {}^B \begin{pmatrix} p_1 \\ p_2 \\ p_3 \end{pmatrix}$$

여기면 frame이 다르므로 계산은 되지만,

meaningful 하지는 않음. 그런데도 매트랩은 이런 계산한다는게 문제. 따라서 이것이 의미는

가치려면 common frame에 있도록 해야함.

$\uparrow$
모두

coordinate frame: $B: \{b_1, b_2, b_3\}$

shorthand notation으로 쓴다.

ex)

$$\begin{pmatrix} 5 \\ 3 \\ 2 \end{pmatrix} : \text{matrix}$$

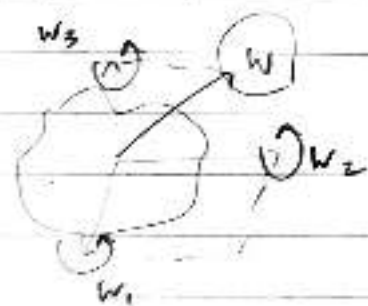
$$E \begin{pmatrix} 5 \\ 3 \\ 2 \end{pmatrix} \equiv (5m)\hat{e}_1 + (3m)\hat{e}_2 + (2m)\hat{e}_3$$

matlab에 vector를 더할 때는 같은 frame이 있도록 해야함. 만약 B, E에 있는 것들은 대하여 rotation matrix로 같은 frame으로 바꾼 후 더함.

Vector $\rightarrow$ matrix (0)

matrix $\rightarrow$ vector (X)

* Angular velocity vector.



$$\vec{w} = w_1 \hat{b}_1 + w_2 \hat{b}_2 + w_3 \hat{b}_3$$

이 rigid body의 $\vec{w}_{net}$ 이 외부 inertial frame N에 대해 상대적으로 어떤 상황인지 알아보는 것이 중요!

이 변환은 자이로스코프가 함.

Angular velocity relative to inertial frame

$$\omega_{B/O} = \omega_{B/N} - \omega_{O/N}$$

(B relative to O) (B relative to N) (O relative to N)

Positive rotation: right hand rule을 이용.

* vector differentiation

What is a derivative of Vector! : 어느 좌표계에서 바라보느냐에 따라 0 or non zero.

$$\dot{\mathbf{r}} = \mathbf{W} \times \mathbf{r}$$

$$\mathbf{r} = r_1 \hat{\mathbf{b}}_1 + r_2 \hat{\mathbf{b}}_2 + r_3 \hat{\mathbf{b}}_3$$

$$\mathbf{W}_{B/N} = \omega_1 \hat{\mathbf{b}}_1 + \omega_2 \hat{\mathbf{b}}_2 + \omega_3 \hat{\mathbf{b}}_3$$

$$\frac{B_d}{dt}(\mathbf{r}) = \dot{r}_1 \hat{\mathbf{b}}_1 + \dot{r}_2 \hat{\mathbf{b}}_2 + \dot{r}_3 \hat{\mathbf{b}}_3 \rightarrow \text{B frame에서 볼 때는 easily calculated}$$

$$\frac{N_d}{dt}(\mathbf{r}) = \dot{r}_1 \hat{\mathbf{b}}_1 + \dot{r}_2 \hat{\mathbf{b}}_2 + \dot{r}_3 \hat{\mathbf{b}}_3 + r_1 \frac{N_d}{dt} \hat{\mathbf{b}}_1 + r_2 \frac{N_d}{dt} \hat{\mathbf{b}}_2 + r_3 \frac{N_d}{dt} \hat{\mathbf{b}}_3$$

$$\dot{\mathbf{r}} = \mathbf{W} \times \mathbf{r} \text{ 이고 } \frac{N_d}{dt}(\hat{\mathbf{b}}_i) = \mathbf{W}_{B/N} \times \hat{\mathbf{b}}_i$$

$$r_1 (\mathbf{W}_{B/N} \times \hat{\mathbf{b}}_1) + r_2 (\mathbf{W}_{B/N} \times \hat{\mathbf{b}}_2) + r_3 (\mathbf{W}_{B/N} \times \hat{\mathbf{b}}_3)$$

$$= \mathbf{W}_{B/N} \times (r_1 \hat{\mathbf{b}}_1 + r_2 \hat{\mathbf{b}}_2 + r_3 \hat{\mathbf{b}}_3)$$

$$= \mathbf{W}_{B/N} \times \mathbf{r}$$

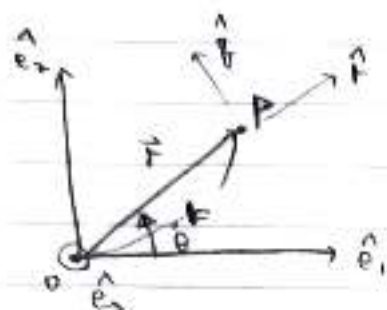
$$\therefore \frac{N_d}{dt}(\mathbf{r}) = \frac{B_d}{dt}(\mathbf{r}) + \mathbf{W}_{B/N} \times \mathbf{r}$$

$$\dot{\mathbf{r}} = \mathbf{W} \times \mathbf{r}$$

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Kinematics: 운동학

Kinetics: 역학



< P를 다루는 방법 >

- ① write vector
- ② define frame
- ③ get W's
- ④ differentiate

$P: \{ \hat{r}, \hat{e}_1, \hat{e}_2 \} \rightarrow$ coordinate frame

$W_{P/E} = \dot{\theta} \hat{e}_3$: P가 Planar로 회전한다고 가정할 시.

inertial frame : non-accelerating frame

$\dot{\vec{r}} \equiv \frac{d}{dt}(\vec{r})$ 그런데 frames을 모아서 P로 바꾸는게 사실 더 편함

$$\frac{d}{dt}(\vec{r}) = \frac{d}{dt}(r\hat{r}) = \dot{r}\hat{r} + r\frac{d}{dt}(\hat{r}) = \dot{r}\hat{r} + r\dot{\hat{r}}$$

$$\dot{\vec{r}} = \frac{d}{dt}(\vec{r}) = \dot{r}\hat{r} + W_{P/E} \times \vec{r}$$

P frame

$$= \dot{r}\hat{r} + \dot{\theta}\hat{e}_3 \times r\hat{r}$$

$$= \dot{r}\hat{r} + r\dot{\theta}\hat{\phi}$$

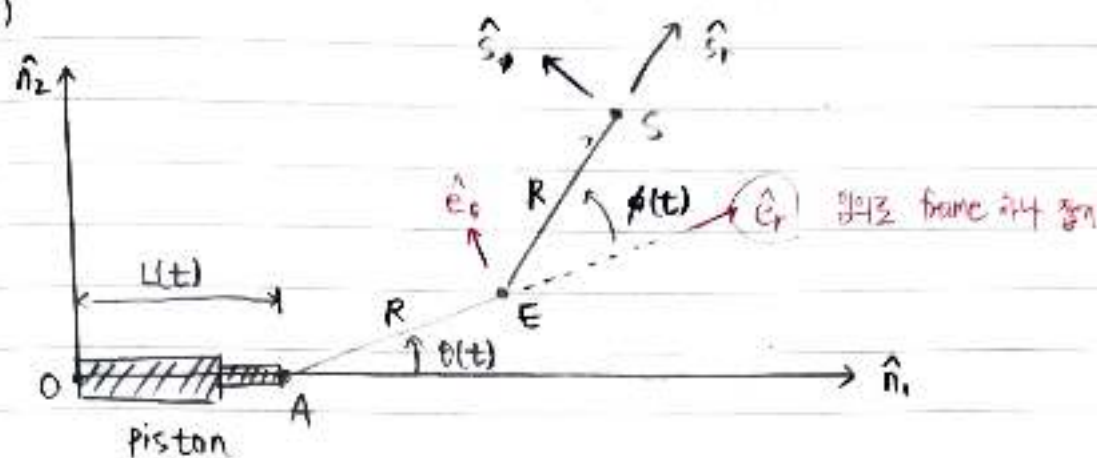
Use rotating frame

angular vel. vector

만약 $\vec{r} = a\hat{a}_1 + b\hat{b}_1$ 라 한다면 가장 lazy하고 easy한 방법으로 풀어나가는 방법은

$$\dot{\vec{r}} = \left\{ \frac{d}{dt}(a\hat{a}_1) + W_{A/N} \times a\hat{a}_1 \right\} + \left\{ \frac{d}{dt}(b\hat{b}_1) + W_{B/N} \times b\hat{b}_1 \right\}$$

ex)



내가 하려고 한 방법

(inertial)
① S의 Velocity를 구하라

Position of A: $L(t) \hat{n}_1$

Position of E: $L(t) \hat{n}_1 +$

$$W_{E/N} = \dot{\theta}$$

$$W_{S/N} = \dot{\theta} + \dot{\phi}$$

각속도는 벡터임!!!

$$\frac{N}{dt} \frac{d}{dt} (\vec{S}) = \frac{E}{dt} \frac{d}{dt} (\vec{S}) + W_{E/N} \times \vec{S}$$

$$\dot{\vec{S}} = \dot{\phi} + \dot{\theta} \times \vec{S}$$

$$\vec{r} = L \hat{n}_1 + R \hat{e}_r + R \hat{s}_r = \vec{r}_{S/N}$$

$$N: \{ \hat{n}_1, \hat{n}_2, \hat{n}_3 \}$$

$$W_{E/N} = \dot{\theta} \hat{n}_3$$

$$E: \{ \hat{e}_r, \hat{e}_\theta, \hat{n}_3 \}$$

$$W_{S/E} = \dot{\phi} \hat{n}_3$$

$$S: \{ \hat{s}_r, \hat{s}_\theta, \hat{n}_3 \}$$

$$W_{S/N} = W_{E/N} + W_{S/E}$$

$$= (\dot{\theta} + \dot{\phi}) \hat{n}_3$$

$$\dot{\vec{r}} = \frac{N}{dt} \frac{d}{dt} (\vec{r}) = \frac{N}{dt} \frac{d}{dt} (L \hat{n}_1) + \left\{ \frac{E}{dt} \frac{d}{dt} (R \hat{e}_r) + W_{E/N} \times (R \hat{e}_r) \right\} + \left\{ \frac{S}{dt} \frac{d}{dt} (R \hat{s}_r) + W_{S/N} \times (R \hat{s}_r) \right\}$$

$$= \dot{L} \hat{n}_1 + \{ \dot{\theta} \hat{n}_3 \times (R \hat{e}_r) \} + \{ (\dot{\theta} + \dot{\phi}) \hat{n}_3 \times (R \hat{s}_r) \}$$

$$= \dot{L} \hat{n}_1 + R \dot{\theta} \hat{e}_\theta + R (\dot{\theta} + \dot{\phi}) \hat{s}_\theta$$

② S의 inertial acceleration은 ?하라

$$\dot{\mathbf{r}} = \dot{L} \hat{\mathbf{n}}_1 + R \dot{\theta} \hat{\mathbf{e}}_\theta + R(\dot{\theta} + \dot{\phi}) \hat{\mathbf{s}}_\phi$$

$$\ddot{\mathbf{r}} = \frac{d}{dt}(\dot{\mathbf{r}})$$

$$= \frac{d}{dt}(\dot{L} \hat{\mathbf{n}}_1) + \frac{d}{dt}(R \dot{\theta} \hat{\mathbf{e}}_\theta) + \frac{d}{dt}(R(\dot{\theta} + \dot{\phi}) \hat{\mathbf{s}}_\phi)$$

$$= \ddot{L} \hat{\mathbf{n}}_1 + \left[\overset{+ R \ddot{\theta} \hat{\mathbf{e}}_\theta}{\omega_{E/N} \times (R \dot{\theta} \hat{\mathbf{e}}_\theta)} \right] + \left[\overset{+ R(\ddot{\theta} + \ddot{\phi}) \hat{\mathbf{s}}_\phi}{\omega_{S/N} \times (R(\dot{\theta} + \dot{\phi}) \hat{\mathbf{s}}_\phi)} \right]$$

$$= \ddot{L} \hat{\mathbf{n}}_1 + R \ddot{\theta} \hat{\mathbf{e}}_\theta - R \dot{\theta}^2 \hat{\mathbf{e}}_r + R(\ddot{\theta} + \ddot{\phi}) \hat{\mathbf{s}}_\phi - R(\dot{\theta} + \dot{\phi})^2 \hat{\mathbf{s}}_r$$

↳ sin, cos로 나눠서 하는 것보다 훨씬 간편, 유용!

③ S위치에서 볼 A위치의 움직임.

$$\mathbf{r}_{A/S} = -R \hat{\mathbf{e}}_r - R \hat{\mathbf{s}}_r$$

very easy

$$\frac{d}{dt}(\mathbf{r}_{A/S}) = -\omega_{E/S} \times R \hat{\mathbf{e}}_r$$

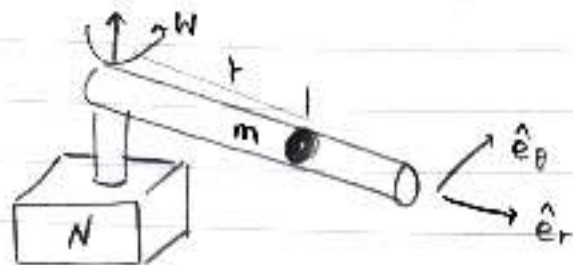
따라서 하는 것

$$= \dot{\phi} \hat{\mathbf{n}}_3 \times R \hat{\mathbf{e}}_r$$

$$= R \dot{\phi} \hat{\mathbf{e}}_\theta$$

<concept check 3> 5/13

1)



$$\vec{r} = r \hat{e}_r \quad \dots \quad (1)$$

$$N: \{ \hat{n}_1, \hat{n}_2, \hat{n}_3 \}$$

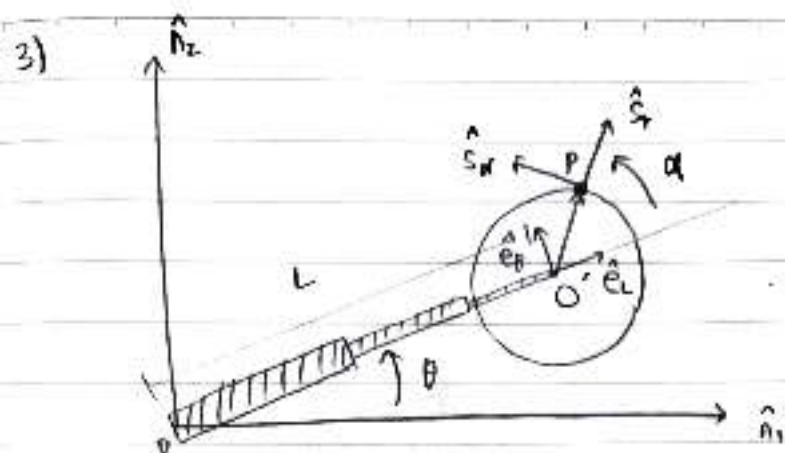
$$E: \{ \hat{e}_r, \hat{e}_\theta, \hat{n}_3 \} \quad \dots \quad (2)$$

$$W_{E/N} = W \hat{n}_3, \quad W = \dot{\theta}$$

$$\begin{aligned} \frac{d}{dt}(\vec{r}) &= \dot{r} \hat{e}_r + W_{E/N} \times r \hat{e}_r \\ &= \dot{r} \hat{e}_r + r W \hat{e}_\theta = \dot{r} \hat{e}_r + r \dot{\theta} \hat{e}_\theta = \dot{\vec{r}} \end{aligned}$$

2) Acceleration

$$\begin{aligned} \frac{d}{dt}(\dot{\vec{r}}) &= \left\{ \frac{d}{dt}(\dot{r} \hat{e}_r) + W_{E/N} \times \dot{r} \hat{e}_r \right\} + \left\{ \frac{d}{dt}(r \dot{\theta} \hat{e}_\theta) + W_{E/N} \times r \dot{\theta} \hat{e}_\theta \right\} \\ &= \left\{ \dot{r} \hat{e}_r + \dot{r} W \hat{e}_\theta \right\} + \left\{ \dot{r} \dot{\theta} \hat{e}_\theta + r \ddot{\theta} \hat{e}_\theta - r W \dot{\theta} \hat{e}_r \right\} \\ &= (\ddot{r} - r \dot{\theta}^2) \hat{e}_r + (2\dot{r} \dot{\theta} + r \ddot{\theta}) \hat{e}_\theta \end{aligned}$$



$$\vec{P} = L\hat{e}_L + r\hat{s}_r \quad \dots \quad (1)$$

$$W_{E/N} = \dot{\theta} \hat{n}_3$$

$$N: \{ \hat{n}_1, \hat{n}_2, \hat{n}_3 \}$$

$$W_{S/E} = \dot{\alpha} \hat{n}_3$$

$$E: \{ \hat{e}_L, \hat{e}_\theta, \hat{n}_3 \}$$

$$W_{S/N} = (\dot{\theta} + \dot{\alpha}) \hat{n}_3 \quad \dots \quad (2)$$

$$S: \{ \hat{s}_r, \hat{s}_\alpha, \hat{n}_3 \} \quad \dots \quad (2)$$

$$\begin{aligned} \frac{d}{dt}(\vec{P}) &= \left\{ \dot{L} \hat{e}_L + W_{E/N} \times L \hat{e}_L \right\} + \left\{ W_{S/N} \times r \hat{s}_r \right\} \\ &= \dot{L} \hat{e}_L + L \dot{\theta} \hat{e}_\theta + r(\dot{\theta} + \dot{\alpha}) \hat{s}_\alpha = \dot{\vec{P}} \end{aligned}$$

4) Acceleration

$$\begin{aligned} \frac{d}{dt}(\dot{\vec{P}}) &= \left\{ W_{E/N} \times \dot{L} \hat{e}_L \right\} + \left\{ (L \ddot{\theta} + \dot{L} \dot{\theta}) \hat{e}_\theta + W_{E/N} \times L \dot{\theta} \hat{e}_\theta \right\} \\ &\quad + \left\{ r(\ddot{\theta} + \ddot{\alpha}) \hat{s}_\alpha + W_{S/N} \times r(\dot{\theta} + \dot{\alpha}) \hat{s}_\alpha \right\} \\ &= \dot{L} \ddot{\theta} \hat{e}_r + (L \ddot{\theta} + \dot{L} \dot{\theta}) \hat{e}_\theta - L \dot{\theta}^2 \hat{e}_L + r(\ddot{\theta} + \ddot{\alpha}) \hat{s}_\alpha - r(\dot{\theta} + \dot{\alpha})^2 \hat{s}_r \\ &= (2\dot{L} \dot{\theta} + L \ddot{\theta}) \hat{e}_\theta - L \dot{\theta}^2 \hat{e}_L + r(\ddot{\theta} + \ddot{\alpha}) \hat{s}_\alpha - r(\dot{\theta} + \dot{\alpha})^2 \hat{s}_r \end{aligned}$$

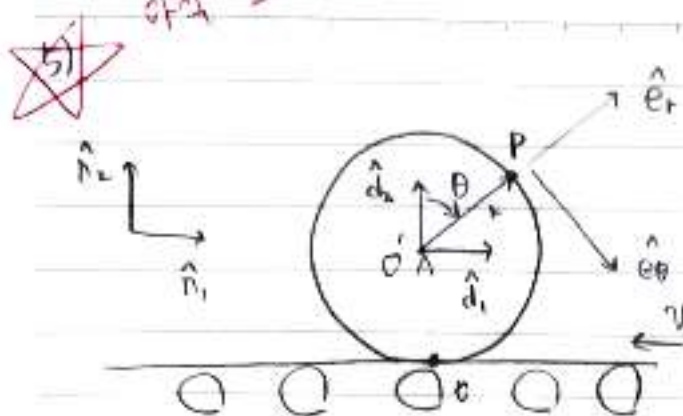
constant rate

아직 모르겠음

inertial acceleration이

어떻게 존재?

↓
좌표계만 inertial, 그 안의 object는
가속도를 가질 수 있음



$$\vec{r} = r\hat{e}_r + vt\hat{d}_1 + r\hat{d}_2$$

$$\omega_{D/N} = 0$$

$$N: \{ \hat{n}_1, \hat{n}_2, \hat{n}_3 \}$$

$$\omega_{E/N} = \omega_{E/D} = -\dot{\theta}\hat{n}_3$$

$$D: \{ \hat{d}_1, \hat{d}_2, \hat{n}_3 \}$$

$$E: \{ \hat{e}_r, \hat{e}_\theta, -\hat{n}_3 \}$$

→ 각도에 따른 회전축이 다르므로 $\hat{e}_3$ 은 $\hat{n}_3$ 과 같지 않음

$$\begin{aligned} \frac{d}{dt}(\vec{r}) &= \left\{ \omega_{E/N} \times r\hat{e}_r \right\} + \left\{ v\hat{d}_1 + \cancel{\omega_{E/N} \times vt\hat{d}_1} \right\} + \left\{ \omega_{D/N} \times r\hat{d}_2 \right\} \\ &= r\dot{\theta}\hat{e}_\theta + v\hat{d}_1 - vt\dot{\theta}\hat{d}_2 \end{aligned}$$



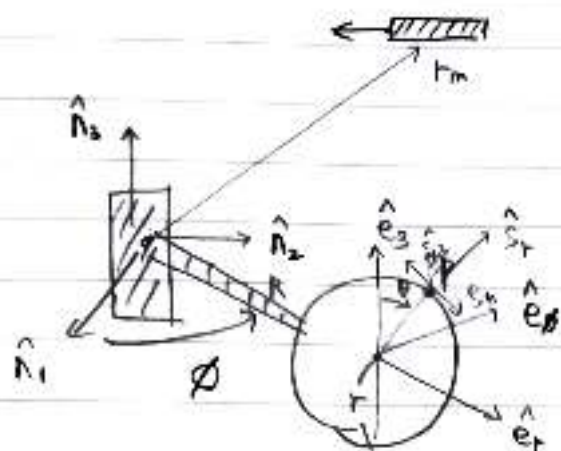
$$\cancel{\omega_{D/N} \times (r\dot{\theta} + v)} - \omega_{E/N} \times r\dot{\theta}\hat{e}_\theta$$

$$= \dot{\theta}\hat{n}_3 \times r\dot{\theta}\hat{e}_\theta = r\dot{\theta}^2\hat{e}_r$$

$$\omega_{D/N} \times (r\dot{\theta} + v)\hat{d}_1 + \omega_{E/N} \times 2r\dot{\theta}\hat{e}_\theta$$

=

n)



$$N: \{ \hat{n}_1, \hat{n}_2, \hat{n}_3 \}$$

$$E: \{ \hat{e}_1, \hat{e}_2, \hat{e}_3 \}$$

$$S: \{ \hat{s}_1, \hat{s}_2, \hat{s}_3 \}$$

$$W_{E/N} = \dot{\phi} \hat{n}_3$$

$$W_{S/E} = \dot{\theta} \hat{s}_3$$

$$W_{S/N} = \dot{\phi} \hat{n}_3 + \dot{\theta} \hat{s}_3$$

$$\vec{p} = R \hat{e}_r + r \hat{s}_r$$

$$\frac{d}{dt}(\vec{p}) = \{ W_{E/N} \times R \hat{e}_r \} + \{ W_{S/N} \times r \hat{s}_r \}$$

$$= R \dot{\phi} \hat{e}_\phi + r \dot{\phi} \sin \theta \hat{s}_3 + r \dot{\theta} \hat{s}_\theta$$

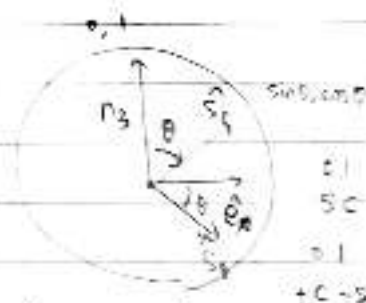
$$b) \{ R \ddot{\phi} \hat{e}_\phi + W_{E/N} \times R \dot{\phi} \hat{e}_\phi \} + \{ (r \ddot{\phi} \sin \theta + r \dot{\theta} \dot{\phi} \cos \theta) \hat{s}_3 + W_{S/N} \times r \dot{\theta} \sin \theta \hat{s}_3 \}$$

$$+ \{ r \ddot{\theta} \hat{s}_\theta + W_{S/N} \times r \dot{\theta} \hat{s}_\theta \}$$

$$= R \ddot{\phi} \hat{e}_\phi - R \dot{\phi}^2 \hat{e}_r + (r \ddot{\phi} \sin \theta + r \dot{\theta} \dot{\phi} \cos \theta) \hat{s}_3$$

$$- r \dot{\phi}^2 \sin \theta \hat{e}_\phi + r \ddot{\theta} \hat{s}_\theta + r \dot{\theta} \dot{\phi} \cos \theta \hat{s}_3 - r \dot{\theta}^2 \hat{s}_r$$

$$= (R \ddot{\phi} - r \dot{\phi}^2 \sin \theta) \hat{e}_\phi + \hat{e}_r (-r \dot{\theta} \dot{\phi} \cos \theta - R \dot{\phi}^2 - r \dot{\phi}^2 \sin \theta - r \dot{\theta} \dot{\phi} \cos \theta) + r \ddot{\theta} \hat{s}_\theta - r \dot{\theta}^2 \hat{s}_r$$



$$\hat{n}_3 \times \hat{s}_r = \hat{s}_3 \times \sin \theta$$

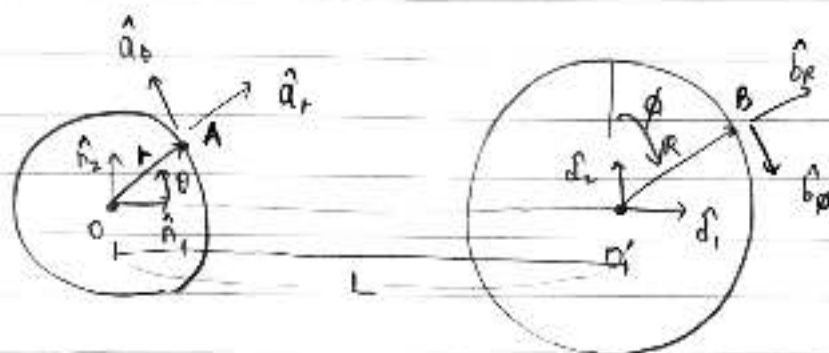
$$9) \quad \vec{r}_m = h \hat{n}_3 - V_0 t \hat{n}_2$$

$$\dot{\vec{r}}_m = -V_0 \hat{n}_2$$

$$\vec{v}_{p/m} = \dot{\vec{r}} = \vec{v}_p - \vec{v}_m$$

$$= R\dot{\phi} \hat{e}_\phi + r\dot{\phi} \sin\theta \hat{s}_3 + r\dot{\theta} \hat{s}_\theta + V_0 \hat{n}_2$$

10)



$$\vec{B} = R\hat{b}_R + L\hat{d}_1, \quad W_{B/N} = -\dot{\phi} \hat{n}_3$$

$$W_{B/N} \times R\hat{b}_R = R\dot{\phi} \hat{b}_\theta = \vec{v}_B$$

$$A: \{ \hat{a}_r, \hat{a}_\theta, \hat{n}_3 \}$$

$$O: \{ \hat{n}_1, \hat{n}_2, \hat{n}_3 \}$$

$$D: \{ \hat{n}_1, \hat{n}_2, \hat{n}_3 \}$$

$$B: \{ \hat{b}_R, \hat{b}_\phi, -\hat{n}_3 \}$$

$$11) \quad W_{B/N} \times R\dot{\hat{b}}_\theta$$

$$= -R\dot{\phi}^2 \hat{b}_R$$

$$W_{B/A} = -\dot{\phi} \hat{n}_3 - \dot{\theta} \hat{n}_3$$

$$12) \quad \vec{A} = r\hat{a}_r, \quad W_{A/N} = \dot{\theta} \hat{n}_3$$

$$\vec{r}_{B/A} = R\hat{b}_R + L\hat{d}_1 - r\hat{a}_r$$

$$\begin{aligned} \frac{d}{dt} (\vec{r}_{B/A}) &= \{ W_{B/A} \times R\hat{b}_R \} + \{ W_{N/A} \times L\hat{d}_1 \} \\ &= R(\dot{\theta} + \dot{\phi}) \hat{b}_\theta - L\dot{\theta} \hat{d}_2 \end{aligned}$$

$$13) \dot{W}_{B/N} = -\dot{\phi} \hat{n}_3$$

$$\overset{N}{\frac{d}{dt}}(-\dot{\phi} \hat{n}_3) = \overset{B}{\frac{d}{dt}}(-\dot{\phi} \hat{n}_3) + \dot{W}_{B/N} \times (-\dot{\phi} \hat{n}_3)$$

☆ ~~모르겠음~~ 해결

5) B frame과 N frame에서 볼 때 항상 같은 것. $\dot{W}_{B/N}$ or $\dot{W}_{N/B}$

Because $\overset{N}{\frac{d}{dt}}(\quad) = \overset{B}{\frac{d}{dt}}(\quad) + \cancel{\dot{W}_{B/N} \times (\quad)}$

cross product 가 0이 되어
앞의 항들만 남기 때문

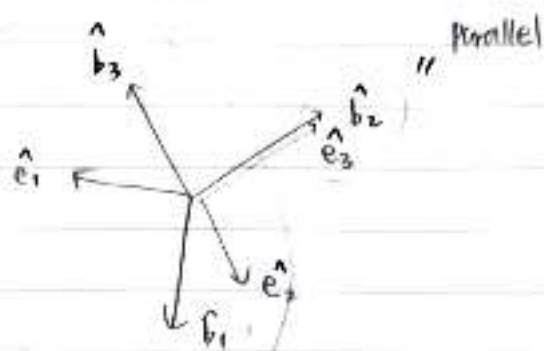
(b)

$$\mathbf{r} = r_1 \hat{b}_1 + r_2 \hat{b}_2 + r_3 \hat{b}_3 \equiv \overset{B}{\begin{pmatrix} r_1 \\ r_2 \\ r_3 \end{pmatrix}}$$

$$\underset{\sim}{\overset{B}{\frac{d}{dt}}}(\mathbf{r}) = \dot{r}_1 \hat{b}_1 + \dot{r}_2 \hat{b}_2 + \dot{r}_3 \hat{b}_3 = \overset{B}{\begin{pmatrix} \dot{r}_1 \\ \dot{r}_2 \\ \dot{r}_3 \end{pmatrix}} = \underset{\sim}{\frac{d}{dt}}(\overset{B}{\mathbf{r}})$$

큰 의미없다.

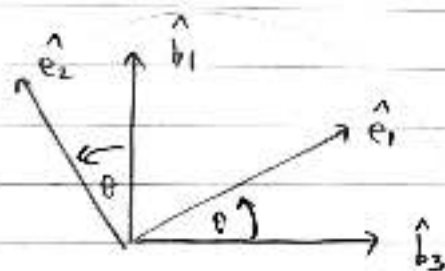
sin과 cos는 회전한 베제하르 transport theorem을 사용한다



b

모두 한 Planar에 존재, 이는 사이는 orthogonal이 하지 않으므로 직접 사용

↓



matrix 사용해서 직접
사용해도 괜찮다

$$\hat{e}_2 = \cos\theta \hat{b}_1 - \sin\theta \hat{b}_3$$

$$\begin{aligned} \hat{e}_2 \times \hat{b}_3 &= (\cos\theta \hat{b}_1 - \sin\theta \hat{b}_3) \times \hat{b}_3 \\ &= -\cos\theta \hat{b}_2 \end{aligned}$$

하나의 방향을 볼